Binomial tree approximation

Consider a risky asset with stock prices S that follows geometric Brownian motion under a risk neutral measure Q. In other words,

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^Q$$

or, equivalently

$$S_t = S_0 e^{(r-\sigma^2/2)t+\sigma W_t^Q},$$

where r is the risk free rate, σ is the volatility and W^Q is a standard Brownian motion under Q. As we have seen in the module, the continuous process S can be approximated by a binary tree. One such approach builds a tree on the interval $[0,T]$ by taking $\Delta t = T/M$, where M is the number of steps in the tree. The steps are equidistant times

$$t_i = i\Delta t \ \text{ for } i = 0,\dots,M,$$

and $S_i^{(M)}$ is the relevant approximation of S_{t_i}. At any time t_i and over a small time interval Δt, the approximation $S_i^{(M)}$ of S_{t_i} is multiplied by u_M or d_M respectively, i.e.

$$S_{i+1}^{(M)} = \begin{cases} S_i^{(M)} u_M & \text{with probability } 0.5 \\ S_i^{(M)} d_M & \text{with probability } 0.5. \end{cases}$$

The drift and volatility parameters of the continuous time process are now captured by the new parameters u_M and d_M, together with the original parameters r and σ. We take $S_0^{(M)} = S_0$ and assume that $u_M > d_M$.

Exercise 2.1 (Binomial tree approximation: 25% of project mark). Design a binary tree method that approximates the Black-Scholes model and can be used to price European options. Your work should include the following:

1. Theoretical work:

   (a) Compute the conditional moments

   $$E^Q\left(S_{(i+1)\Delta t}^k \middle| S_{i\Delta t} = S\right)$$

   in the continuous-time model and

   $$E^Q\left(\left(S_{i+1}^{(M)}\right)^k \middle| S_i^{(M)} = S\right)$$

   in the binomial tree model, where $k = 1, 2$.
   For $k = 1$:

   $$\begin{aligned} E^Q(S_{(i+1)\Delta t}|S_{i\Delta t} = S) &= Se^{(r-\frac{1}{2}\sigma^2)\Delta t} E^Q_{i\Delta t}\left[e^{\sigma\left(W^Q_{(i+1)\Delta t} - W^Q_{i\Delta t}\right)}\right] \\ &= Se^{(r-\frac{1}{2}\sigma^2)\Delta t} E^Q\left[e^{\sigma\sqrt{\Delta t}Z}\right] \\ &= Se^{(r-\frac{1}{2}\sigma^2)\Delta t} e^{\frac{1}{2}\sigma^2\Delta t} \\ &= Se^{r\Delta t - \frac{1}{2}\sigma^2\Delta t + \frac{1}{2}\sigma^2\Delta t} \\ &= Se^{r\Delta t} \end{aligned}$$

   and

   $$E^Q\left(S_{i+1}^{(M)} \middle| S_i^{(M)} = S\right) = \frac{1}{2}S\left(u_M + d_M\right)$$

For $k=2$:

$$\begin{aligned}
E^Q(S^2_{(i+1)\Delta t}|S_{i\Delta t}=S) &= S^2 e^{2(r-\frac{1}{2}\sigma^2)\Delta t} E^Q_{i\Delta t}\left[e^{2\sigma\left(W^Q_{(i+1)\Delta t}-W^Q_{i\Delta t}\right)}\right] \\
&= S^2 e^{2(r-\frac{1}{2}\sigma^2)\Delta t} E^Q\left[e^{2\sigma\sqrt{\Delta t}Z}\right] \\
&= S^2 e^{2(r-\frac{1}{2}\sigma^2)\Delta t} e^{2\sigma^2\Delta t} \\
&= S^2 e^{2r\Delta t-\sigma^2\Delta t+2\sigma^2\Delta t} \\
&= S^2 e^{(2r+\sigma^2)\Delta t}
\end{aligned}$$

and

$$E^Q\left(\left(S^{(M)}_{i+1}\right)^k \middle| S^{(M)}_i = S\right) = \frac{1}{2}S^2\left(u_M^2+d_M^2\right)$$

(b) Given the values of M and Δt, derive formulae for u_M and d_M in terms of r and σ by matching the first and second order conditional moments of the continuous-time model and the binomial tree model. *Hint: use the quadratic formula* $x=\frac{1}{2a}(-b\pm\sqrt{b^2-4ac})$ for the solutions of the quadratic equation $ax^2+bx+c=0$.

By equating the first two conditional moments, we get the following equations:

$$\begin{aligned}
u_M + d_M &= 2e^{r\Delta t} && (1) \\
u_M^2 + d_M^2 &= 2e^{(2r+\sigma^2)\Delta t} && (2)
\end{aligned}$$

We know that $(u_M+d_M)^2 = u_M^2+d_M^2+2u_Md_M$ and so can substitute equations (1) and (2) to get

$$4e^{2r\Delta t} = 2e^{(2r+\sigma^2)\Delta t} + 2\gamma$$

if we let $u_Md_M=\gamma$ then we can solve for γ to find:

$$\gamma = 2e^{2r\Delta t} - e^{(2r+\sigma^2)\Delta t} = e^{2r\Delta t}\left(2-e^{\sigma^2\Delta t}\right)$$

Now as $\gamma=u_Md_M$ we can substitute $d_M=\gamma/u_M$ into equation (1) to get

$$\begin{aligned}
u_M + \frac{\gamma}{u_M} &= 2e^{r\Delta t} \\
u_M^2 - 2u_Me^{r\Delta t} + \gamma &= 0
\end{aligned}$$

Using the quadratic formula, we can solve this to give

$$u_M = \frac{1}{2}\left[2e^{r\Delta t} \pm \sqrt{4e^{2r\Delta t}-4\gamma}\right] = e^{r\Delta t}\left(1\pm\sqrt{e^{\sigma^2\Delta t}-1}\right)$$

Similarly for d_M we find

$$d_M = e^{r\Delta t}\left(1\pm\sqrt{e^{\sigma^2\Delta t}-1}\right)$$

As, by definition, $d_M<u_M$ we choose the positive root for u_M and the negative root for d_M then

$$u_M = e^{r\Delta t}\left(1+\sqrt{e^{\sigma^2\Delta t}-1}\right), \qquad d_M = e^{r\Delta t}\left(1-\sqrt{e^{\sigma^2\Delta t}-1}\right)$$

2. State the algorithm used for approximating the price of a European-style derivative security with payoff at time T given by $f(S_T)$ by means of this tree approximation.

   Approximating European-style option price with payoff:

   - Input:
     - Black-Scholes parameters: S_0, r, σ, T
     - Path independent option, payoff function $f(S_T)$ and its arguments
     - Number of steps: M
   - Parameters:
     - $\Delta t = T/M$
     - $u_M = e^{r\Delta t}\left(1 + \sqrt{e^{\sigma^2 \Delta t} - 1}\right)$
     - $d_M = e^{r\Delta t}\left(1 - \sqrt{e^{\sigma^2 \Delta t} - 1}\right)$
   - Output:
     - Initial price V

   i) Choose M and calculate $\Delta t = T/M$.

   ii) Calculate $u_M = e^{r\Delta t}\left(1 + \sqrt{e^{\sigma^2 \Delta t} - 1}\right)$ and $d_M = e^{r\Delta t}\left(1 - \sqrt{e^{\sigma^2 \Delta t} - 1}\right)$

   iii) Calculate Terminal Stock Prices: $S_0 u_M^j d_M^{M-j} = S_{M,j}$

   iv) Calculate Terminal Payoff: $f(S_{M,j}) = P_{M,j}$

   v) Set the Terminal Price: $V_{M,j} = P_{M,j}$

   vi) For $i = 0, \ldots, M-1$, do the following:

      a) $V_{m,j} = \frac{1}{2}e^{-r\Delta t}(V_{m+1,j+1} + V_{m+1,j})$

5. Write a few paragraphs (no more than 200 words) to compare this tree approximation to the one that was covered in the lectures. Which method would you recommend, and why?

   Both the model from lectures and the model used in this problem are binomial tree methods to approximate the option price. Both will approximate the payoff by working backwards through a recombinant tree with $O(M^2)$. Additionally, both methods converge to the Black-Scholes price as $M \to \infty$. We note that the model derived in this problem has pre-determined probabilities for the up and down step, causing our γ to be determined by the time-step, risk-free rate and volatility. We then must use this $\gamma = u \times d$ to calculate the up and down step, whereas in the lecture model we can anchor γ causing the method to have smoother convergence. We do this by allowing it to be a function of our initial stock, S_0 and strike price, K, ensuring that the median grid point falls on K. I would recommend that the lecture method be used over the derived method because of the flexibility in anchoring γ.